JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

III Semester B.E Degree Examination

Department of Mathematics

ENGINEERING MATHEMATICS-III

(Common to CSE,ISE)

Duration: 3 Hours | Max. Marks: 100

Instructions:

1. Answer all the questions in PART – A.
2. Questions in PART – B have internal choice.

PART – A

Q.NO	1	CO	1	CD	L2	PI	1.1.1	MARKS	10

Fit a curve of the form $y = ax^2 + bx + c$ *for* the following data, also estimate y when x = 6

x	1	2	3	4	5
y	10	12	13	16	19

Q.NO	2	CO	2	CD	L3	PI	1.3.1	MARKS	10

(a) There are 10 students of which 3 are graduates. If a committee of 5 is to be formed, what is the probability that there are (i) only 2 graduates (ii) at least 2 graduates.

(b) A bag contains three coins one of which is two headed and other two are normal and fair. A coin is chosen at random from the bag and tossed four times in succession. If head turns up each time, what is the probability that this is a two headed coin.

Q.NO	3	CO	3	CD	L2	PI	1.2.1	MARKS	10

(a)The discrete probability distribution of X is given below:

X	0	1	2	3	4	5	6	7
$P(x)$	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

(i) Find the value of k ii) Evaluate $P(x < 6)$, $P(3 < x \leq 6)$

(b)The probability that a news reader commits no mistake in reading the news is $1/e^3$. Find the probability that on a particular news broadcast he commits (i) only 2 mistakes (ii) at most 3 mistakes

USN													Course Code:	

Q.NO	4	CO	4	CD	L1	PI	1.2.1	MARKS	10

(a) How many permutations are there for the eight letters *a*, *c*, *f*, *g*, *i*, *t*, *w*, *x*?

(i) How many starts with the letter *f*? (ii) How many letters start with *t* and end with the letter *c*?

(b) How many positive integers *n* can we form using the digits 3, 4, 4, 5, 5, 6, 7 if we want *n* to exceed 5,000,000?

Q.NO	5	CO	5	CD	L2	PI	1.2.1	MARKS	10

Five teachers T_1, T_2, T_3, T_4, T_5 are to be made class teachers for five classes C_1, C_2, C_3, C_4, C_5, one teacher for each class. T_1 and T_2 do not wish to become the class teachers for C_1 or C_2, T_3 and T_4 for C_4 or C_5, and T_5 for C_3 or C_4 or C_5. In how many ways can the teachers be assigned the work (without displeasing any teacher)

PART – B

Q.NO	6	CO	1	CD	L3	PI	1.2.1	MARKS	10

Fit an exponential curve of the form $y = ae^{bx}$ for the following data.

x	5	6	7	8	9	10
y	133	55	23	7	2	2

OR

Q.NO	7	CO	1	CD	L3	PI	1.2.1	MARKS	10

Compute the coefficient of correlation and hence obtain the equation of the lines of regression for the following data

x	1	2	3	4	5
y	2	5	3	8	7

Q.NO	8	CO	2	CD	L2	PI	1.3.1	MARKS	10

(a)State and prove Baye's theorem.

(b) Three major parties *A*, *B C* are contending for power in the elections of a state and the chance of their winning the election is in the ratio 1 : 3 : 5. The parties A, *B C* respectively have probabilities of banning the online lottery 2/3, 1/3, 3/5. What is the probability that there will be a ban on the online lottery in the state? What is the probability that the ban is from the party *C*?

OR

USN													Course Code:	22MA312/24MA313

Q.NO	9	CO	2	CD	L1	PI	1.2.1	MARKS	10

(a) If A and B are independent events, show that the events
(i) $\bar{A}$ and $\bar{B}$ are also independent. (ii) A and $\bar{B}$ are also independent.

(iii) $\bar{A}$ and B are also independent

(b) If A and B are events with $P(A) = \frac{1}{2}, P(A \cup B), = \frac{3}{4}$ *and* $P(\bar{B}) = \frac{5}{8}$, then find $P(A \cap B)$, $P(\bar{A} \cap \bar{B}), P(\bar{A} \cup \bar{B})$, $P(\bar{A} \cap B)$

Q.NO	10	CO	3	CD	L2	PI	1.2.1	MARKS	10

(a) From a sealed box containing a dozen apples it was found that 4 apples are perished. Obtain the probability distribution of the number of perished apples when 3 apples are drawn at random.

(b) In 800 families with five children each how many families would be expected to have (i)either 2 or 3 Boys (ii)at most 2 Girls.

OR

Q.NO	11	CO	3	CD	L3	PI	1.2.1	MARKS	10

(a) A continuous random variable has the probability density function

$f(x) = \begin{cases} kx^2, 0 < x < 3 \\ 0, \quad Otherwise \end{cases}$ (i) Find the value of k ii) Evaluate P(1< x < 2)),

(b) If x is normal variate with mean 30 and standard deviation 5 find the probability that

(i) 26 ≤ x ≤ 40) (ii) x ≥ 45, Given $\phi(0.8)$=0.2881, $\phi(2) = 0.4772$, $\phi(3) = 0.4987$

where $\phi(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{\frac{-z^2}{2}} dz$

Q.NO	12	CO	4	CD	L1	PI	1.2.1	MARKS	10

(a) Find the coefficient of $w^2 x^2 y^2 z^2$ in the expansion of $(2w - x + 3y + z - 2)^{12}$

(b) Determine the number of positive Integers n such that $1 \leq n \leq 100$ and n is not divisible by 2, 3 or 5.

OR

Q.NO	13	CO	4	CD	L1	PI	1.2.1	MARKS	10

(a)Determine in how many ways can the letters in **ARRANGEMENT** be arranged so that (i) there are exactly two pairs of consecutive identical letters (ii) at least two pairs of consecutive identical letters.

USN													Course Code:	22MA312/24MA313

Q.NO	14	CO	5	CD	L1	PI	1.3.1	MARKS	10
Find the number of derangements of 1, 2, 3, 4?									
OR									
Q.NO	15	CO	5	CD	L3	PI	1.3.1	MARKS	10
Using the exponential generating function, find the number of ways in which four of the letters in the word **ENGINE** be arranged									

Course Outcome (CO):	
CO 1	Apply least square method to fit a curve for the given data and evaluate the correlation coefficient and regression lines for the data.
CO 2	Determine the nature of the events and hence calculate the appropriate probabilities of the events.
CO 3	Classify the random variables to determine the appropriate probability distribution.
CO 4	Apply the techniques of counting to identify the number of ways in which a given task can be accomplished without list all the possibilities explicitly
CO 5	Derive the generating function for the given situation and evaluate the required coefficient.

Performance Indicators:	
1.1.1	Apply mathematical techniques such as calculus ,linear algebra and statistics to solve problems.
1.2.1	Apply laws of natural science to an engineering problem
1.3.1	Apply fundamental engineering concepts to solve engineering problem

Cognitive Domain (CD):	
L1	Remember
L2	Understand
L3	Apply
L4	Analyze
L5	Evaluate
L6	Create

USN	0	1	J	S	T	2	4	0	C	S	1	2	Code:	

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Third Semester ,B E Degree Examination

Department of Computer Science and Engineering

DATASTRUCTURES

Duration: 3 Hours | Max. Marks: 100

Instructions:

- All Questions in Part A is compulsory.
- Part B has internal choice, Answer any one full question.

PART – A

Q.NO	1	CO	1	CD	L2	PI	1.4.1	MARKS	10

Define Data structure. Explain the classification of Data structures with suitable examples.

Q.NO	2	CO	2	CD	L2	PI	2.2.4	MARKS	5+5=10

a) Demonstrate the difference between Single and doubly linked list with examples.
b) Explain the significance of header node in linked list with example.

Q.NO	3	CO	3	CD	L3	PI	3.2.2	MARKS	10

Apply expression conversion algorithm and convert the following given infix expression into prefix as well as postfix expression.(Indicate the stack content after each step)

((A – B) + C) $ (D + (E – F))

Q.NO	4	CO	4	CD	L2	PI	2.1.2	MARKS	10

Explain the following with suitable examples.

a) Complete Binary tree
b) Binary search tree
c) Balanced Binary tree
d) Skewed Binary tree

Q.NO	5	CO	5	CD	L2	PI	2.1.2	MARKS	10

Discuss the different storage structures of graphs with suitable examples.

PART – B

Q.NO	6	CO	1	CD	L3	PI	3.2.2	MARKS	5+5

a) Develop a function in C language that checks whether given two arrays are identical

or not using pointers.

b) Develop a C program using pointers to read array of n elements and display the content of the array in reverse order using pointers.

OR

Q.NO	7	CO	1	CD	L3	PI	3.2.2	MARKS	10

Define a Structure Employee with the following details name, employee ID, department name, salary and years of experience. Develop a C program to read the employee details, calculates the salary hike based on years of experience and display the details of employees of a given department.

Years of Experience	Salary hike
< 5years	5%
5 years to 10 years	10%
Above 10 years	15%

Q.NO	8	CO	2	CD	L2	PI	3.2.2	MARKS	10

Define Circular linked list and Develop C functions of circular single linked list to perform the following operations

i) count the number occurrences of key value
ii) reverse the list

OR

Q.NO	9	CO	2	CD	L3	PI	3.2.2	MARKS	10

Define Doubly linked list and Develop C functions of circular doubly linked list to perform the following operations

i) Insert at rear
ii) Search for an item.

Q.NO	10	CO	3	CD	L3	PI	4.1.2	MARKS	10

Explain the applications of Queue. Develop C functions to perform the following queue operations

i) Enqueue or insertion
ii) Dequeue or deletion

OR

USN											Code:	24CS330

Q.NO	11	CO	3	CD	L3	PI	4.1.2	MARKS	10
Define Tower of Hanoi problem. Provide step wise solution to solve the same with 3 discs. Develop a C function to solve Tower of Hanoi problem.									

Q.NO	12	CO	4	CD	L3	PI	4.1.2	MARKS	5+5
Develop C functions to perform the following: i) To find the minimum node in Binary Search Tree (BST). ii) Develop a C function to insert a node into Binary Search Tree.									

OR

Q.NO	13	CO	4	CD	L3	PI	4.1.2	MARKS	5+5
Define Binary tree. Construct a binary tree by considering given traversals and also provide the post order traversal for the same. Pre-order: ABCGHDEFI In-order: GCHBDAEIF									

Q.NO	14	CO	5	CD	L3	PI	4.1.2	MARKS	5+5
a) Differentiate between i) Directed and Undirected graphs. ii) Connected and disjoint graphs b) Define the following with respect to graphs: i) Graph ii) path iii) in-degree of a vertex iv) out-degree of a vertex v) cycle									

OR

Q.NO	15	CO	5	CD	L3	PI	4.1.2	MARKS	5+5
Construct i) BFS(Breadth First Search) tree by performing the BFS traversal ii) DFS(Depth First search) tree by performing DFS traversal from vertex A for the following graph:									

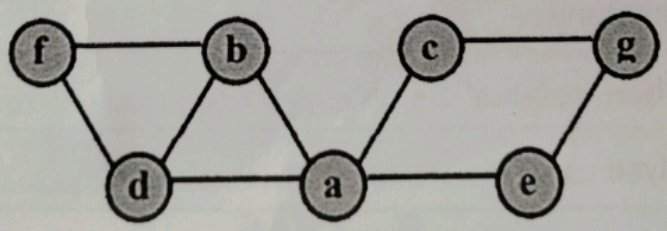

Figure : Q15

USN	0	1	J	S	T	2	4	U	C	S	1	2	0

24CS320

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JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

III Semester B.E Degree Examination

Department of Computer Science and Engineering

DIGITAL SYSTEM DESIGN

Duration: 3 Hours | Max. Marks:100

Instructions:

- PART- A Questions are Compulsory.
- PART- B Questions are having internal choice.

PART – A

Q.NO	CO	CD	PI	MARKS
1.	CO1	L3	1.3.1	4+6
	a. Simplify, $F = (X+ Y)\ (X'(Y' + Z'))' + X'(Y + Z)$ using Boolean laws and theorems. b. Write a note on conversion between SOP and POS considering a three variable truth table.			
2.	CO2	L3	1.4.1	10
	Show how 4 variable truth table can be realized using an 8-to-1 multiplexer.			
3.	CO3	L2	2.3.2	10
	Explain the working of NOR-Gate Latch and NAND-Gate Latch with logic symbol and truth table.			
4.	CO4	L3	2.2.3	10
	Explain the working of Ring counter and Switched Tail counter with diagrams.			
5.	CO5	L3	2.3.2	10
	Design a Mod-8 Synchronous Counter using J-K flipflops.			

PART – B

Q.NO	CO	CD	PI	MARKS
6.	CO1	L3	1.4.1	10
	Design a system which accepts four inputs a,b,c,d and outputs a 1 whenever the least significant input or the most significant input goes high. Use Karnaugh map to simplify the expression and design the circuit.			
	OR			
7.	CO1	L3	2.2.3	10
	For the following Boolean function, use the Quine-McClusky method to obtain all the Prime Implicants. $F(a,b,c,d) = \Sigma m(7,9,12,13,14,15)+dc(4,11)$.			
8.	CO2	L2	1.4.1	10
	With logic diagram and truth table explain BCD to Decimal decoder.			
	OR			
9.	CO2	L2	1.3.1	10
	With required diagrams, describe parity generators and checkers.			
10.	CO3	L2	2.2.3	10
	Explain JK Master Slave flipflop with required diagrams.			
	OR			
11.	CO3	L2	2.2.4	10
	Write a note on conversion of flip flops.			
12.	CO4	L2	2.2.4	10
	Discuss in detail about Shift registers types with diagrams.			
	OR			
13.	CO4	L2	2.2.4	5+5
	Discuss with logic diagram (i) Serial in Serial Out (SISO) (ii) Parallel in and Parallel Out (PIPO) registers.			

14.	CO5	L2	2.3.2	10
	What are Asynchronous Counters? Explain three bits asynchronous down counter with logic circuit, truth table and waveforms.			
	OR			
15.	CO5	L2	2.2.4	10
	Explain two distinct models by which a synchronous sequential logic circuit can be designed with state transition diagrams and state synthesis table.			

Course Outcome (CO):	
CO–1	Explain the fundamental concepts of Boolean algebra, Boolean theorems, K-MAP, Quine-McCluskey algorithm and apply the knowledge of theoretical foundations to simplify and solve Boolean functions.
CO–2	Design and implement the combinational circuits.
CO–3	Apply the knowledge of timing aspects and operation of Flip-Flops in conversion between different Flip Flops.
CO–4	Illustrate the working of various registers and register circuits.
CO–5	Design and implement synchronous/asynchronous counters and synchronous sequential circuits.

PI	
1.3.1	Apply Engineering Fundamentals.
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem.
2.2.3	Identify existing solution/methods to solve the problem, including forming justified approximations and assumptions.
2.2.4	Compare and contrast alternative solution/methods to select the best methods.
2.3.2	Identify design constraints for required performance criteria.

CD		
L1 - Remember	L2 - Understand	L3 - Application

--- End ---

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

III Semester BE Degree Examination

Department of Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING USING C++

Duration: 3 Hours | Max. Marks: 100

Instructions:

- All questions in PART-A is compulsory.
- Part-B has internal choice.

PART – A

Q.NO	1	CO	1	CD	L2	PI	1.3.1	MARKS	10

Explain any five features of Object-Oriented Programming (OOP) in detail with suitable examples.

Q.NO	2	CO	2	CD	L2	PI	1.3.1	MARKS	10

Explain the different types of constructors in C++ with suitable examples for each. Describe how the compiler invokes each type of constructor.

Q.NO	3	CO	3	CD	L2	PI	1.4.1	MARKS	10

Explain the various types of inheritance in C++. Illustrate each type with a neat diagram and an example program.

Q.NO	4	CO	4	CD	L2	PI	1.4.1	MARKS	10

With a neat diagram, explain input and output data streams. Demonstrate the unformatted I/O member functions with a C++ program.

Q.NO	5	CO	5	CD	L2	PI	1.1.1	MARKS	10

What is a template? Explain two types of templates supported by C++ with suitable examples for each.

PART – B

Q.NO	6	CO	1	CD	L2	PI	1.3.1	MARKS	5+5

a) Write a C++ program to rotate the values of three variables using reference variables such that: x takes the value of y, y takes the value of z and z takes the value of x.

b) Explain inline function in C++ with an example program and show the output.

OR

Q.NO	7	CO	1	CD	L2	PI	1.3.1	MARKS	5+5

a) What are default arguments? Write a C++ program that uses a class with a member function that demonstrates default arguments with sample output.
b) Explain dynamic memory allocation and deallocation for an integer with an example program.

Q.NO	8	CO	2	CD	L3	PI	1.1.1	MARKS	5+5

a) How does operator overloading help in converting basic type to class type and class type to basic type? Explain with a sample code.
b) Write a C++ program to create a class Student with data members - id and marks. Overload the == operator using a member function to compare if two students have the same marks.

OR

Q.NO	9	CO	2	CD	L3	PI	1.1.1	MARKS	5+5

a) Describe the structure of a class in C++. Explain how member functions can be defined both inside and outside a class with a sample code.
b) Write a C++ program to create two classes. Define a friend function to access the private members of both classes and display their sum.

Q.NO	10	CO	3	CD	L3	PI	1.4.1	MARKS	10

Write a C++ program to demonstrate a virtual function. Create a base class Shape with a virtual display () function. Derive classes Rectangle and Circle from it, each providing its own display () function to show a suitable message. Use a base class pointer to call the display () function and demonstrate runtime polymorphism.

OR

Q.NO	11	CO	3	CD	L3	PI	1.4.1	MARKS	5+5

a) Explain function overriding in C++ with an example. How does it differ from function overloading?
b) Explain the order of execution of constructors when class C inherits from class A and virtual class B in public mode and an object of class C is created. Illustrate your answer with a C++ program and show the output.

Q.NO	12	CO	4	CD	L2	PI	1.3.1	MARKS	5+5

a) Write a C++ program to create a file named "ITEM", store item name and cost in it, and then read and display the stored data. Use constructor functions to open the file.
b) Explain the following ios format functions with sample code and output: width(), precision(), fill(), setf(),unsetf().

OR

Q.NO	13	CO	4	CD	L2	PI	1.3.1	MARKS	5+5
a) Explain exception handling mechanism with an example program. b) Explain user-defined manipulators in C++ with syntax and an example program. Also write the output of the program.									

Q.NO	14	CO	5	CD	L3	PI	1.1.1	MARKS	10
Write a C++ program to create a class template Queue. Implement insert_front and delete_rear and display as function templates inside the class. Use this class template to implement a linear queue of integers and a linear queue of characters. Demonstrate the queue operations by displaying the contents.									
OR									
Q.NO	15	CO	5	CD	L3	PI	1.1.1	MARKS	10
Write a C++ program to create a class template Stack with push, pop and display member functions. Use this template to implement a stack of integers and a stack of characters. Demonstrate the stack operations by displaying the contents.									

Course Outcome (CO):	
CO–1	Understand object-oriented programming concepts and be able to design solutions for a given problem.
CO–2	Able to reuse code with extensible Class types, user defined operators and function overloading.
CO–3	Achieve code reusability and extensibility by means of Inheritance and Polymorphism.
CO–4	Incorporate file-handling functions and exceptions for a given application.
CO–5	Design C++ packages with templates for providing solutions to complex problems.

PI	
1.1.1	*Apply the knowledge of discrete structures, linear algebra, statistics and numerical techniques to solve problems*
1.3.1	*Apply engineering fundamentals*
1.4.1	*Apply theory and principles of computer science and engineering to solve an engineering problem*

USN	0	1	J	S	T	2	4	U	C	S	1	2	0	Course Code:	24CS350

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

III Semester BE Degree Examination

Department of Computer Science and Engineering

DISCRETE MATHEMATICAL STRUCTURES

Duration: 3 Hours | Max. Marks: 100

Instructions:

- Part A is Compulsory
- Part B has internal choice.

PART – A

Q.NO	1	CO	1	CD	L3	PI	1.2.1	MARKS	5+5=10

a) Define logical eqivalence. Prove that for any propositions p,q,r the compound proposition

$[p \rightarrow (q \wedge r)] \Leftrightarrow [(p \rightarrow q) \wedge (p \rightarrow r)]$ using truth table.

b) Test the validity of the following argument using rules if inference.

$[(p \rightarrow r) \wedge (q \rightarrow r)] \Longrightarrow (p \vee q) \rightarrow r$

Q.NO	2	CO	2	CD	L3	PI	1.2.1	MARKS	10

Let A = {1,2,3,4,6,8,12}. On A, define the partial ordering relation by xRy if and only if x *divides* y

i) Prove that R is a partial order on A.
ii) Write the Relation matrix.
iii) Draw the Hasse diagram.

Q.NO	3	CO	3	CD	L3	PI	2.6.3	MARKS	5+5

a) For each of the following encoding functions, find the minimum distance between the code words. Indicate the error detecting and error correcting capabilities of each code.

$E : Z_2^2 \rightarrow Z_2^5$ defined by E(00) = 00001, E(01) = 01010, E(10) = 10100, E(11) = 11111

b) The word c = 1010110 is transmitted through a binary symmetric channel. If p = 0.02 is the probability of incorrect receipt of a signal, find the probability that c is received r = 1011111. Determine the error pattern.

Q.NO	4	CO	4	CD	L2	PI	1.7.1	MARKS	10

Define the terms with example:

i) Simple graph

USN | | | | | | | | | | | | Code:

ii) Multigraph
iii) Planar graph.
iv) Bipartite graph
v) Complete Bipartite graph

Q.NO	5	CO	5	CD	L3	PI	1.7.1	MARKS	10

Construct an optimal prefix code for the message ROAD IS GOOD. Decode Message.

PART – B

Q.NO	6	CO	1	CD	L3	PI	1.2.1	MARKS	6+4

a) Test the validity of the following argument using rules of inference.

If i study, i will not fail in the examination.
If i do not watch TV in the evenings, i will study
I failed in the examination.
Therefore i might have watched TV in the evenings

b) For the universe of all integers, Let
P(x) : x > 0 **Q(x)** : x is even **R(x)** : x is a perfect square
S(x) : x is divisible by 3 **T(x)** : x is divisible by 7.
Write down the following quantified statements in symbolic form.

i) There exists a positive integer that is even.
ii) Some even integers are divisible by 3.
iii) Every integer is either even or odd.
iv) If x is even and a perfect square, then s is divisible by 3.

OR

Q.NO	7	CO	1	CD	L3	PI	1.2.1	MARKS	6+4

a) Establish the validity of the following argument using rules of inference.

$(\neg p \vee q) \rightarrow r$

$r \rightarrow (s \vee t)$

$\neg s \wedge \neg v$

$\neg u \rightarrow \neg t$

$\therefore p$

b) If $p(x) = x \geq 0$, $q(x)$: $x^2 \geq 0$, $r(x)$: $x^2 - 3x - 4 = 0$, $s(x)$: $x^2 - 3 > 0$, Find the truth values of the following if the universe contains set of real numbers.

i) $\exists x,[p(x) \wedge q(x)]$
ii) $\forall x,[p(x) \rightarrow q(x)]$
iii) $\forall x,[q(x) \rightarrow s(x)]$
iv) $\forall x,[r(x) \vee s(x)]$

USN													Course Code:	24CS350

Q.NO	8	CO	2	CD	L3	PI	1.7.1	MARKS	10

Let A={1,2} B={m,n,p} C={3,4,5}. Consider relation $R_1 \subseteq (A \times B)$, $R_2 \subseteq (B \times C)$ and $R_3 \subseteq (B \times C)$ defined by

$R_1 = \{(1,m), (1,n), (1,p)\}$

$R_2 = \{(m,3), (m,4), (n,3)\}$

$R_3 = \{(n,3), (n,5), (p,4)\}$

Determine the following

i) $R_1 \circ (R_2 \cup R_3)$
ii) $R_1 \circ (R_2 \cap R_3)$
iii) $(R_1 \circ R_2) \cup (R_1 \circ R_3)$
iv) $(R_1 \circ R_2) \cap (R_1 \circ R_3)$

OR

Q.NO	9	CO	2	CD	L3	PI	1.71	MARKS	10

Which elements of the POSET ({2,4,5,10,12,20,25} /) are maximal and which are minimal.

Q.NO	10	CO	3	CD	L3	PI	2.6.3	MARKS	10

A encoding function E: $z_2^2 \rightarrow z_2^5$ is given by generator matrix

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

i) Determine all the code words. What can be said about the error detection capability of this code. What about its error correction capability?

ii) Find the associated parity check matrix H.

iii) Use H to decode the received words 11101, 11011

OR

Q.NO	11	CO	3	CD	L3	PI	2.6.3	MARKS	5+5=10

The generator matrix for an encoding function E: $z_2^3 \rightarrow z_2^6$ is given by

$$G = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}$$

a) Find the code words assigned to 110 and 010

b) Obtain associated parity-check matrix

c) Hence decode the received words: 110110 and 111101

d) Show that the decoding of 111111 is not possible by using H

Q.NO	12	CO	4	CD	L2	PI	2.6.3	MARKS	10

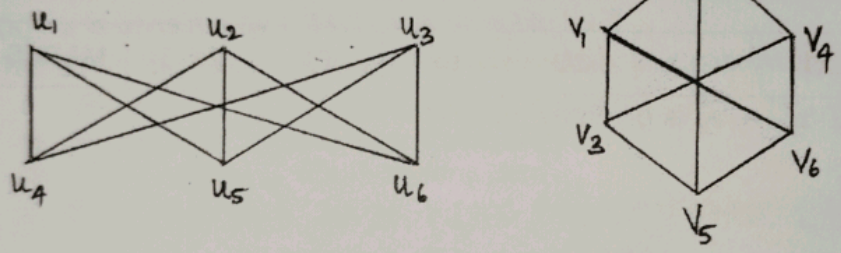
Define the terms with example

i) Walk
ii) Euler Trail
iii) Euler Circuit
iv) Hamilton Path
v) Hamilton cycle.

OR

Q.NO	13	CO	4	CD	L3	PI	2.6.3	MARKS	10

Define isomorphism. Prove that the two graphs shown in the figure below are isomorphic.



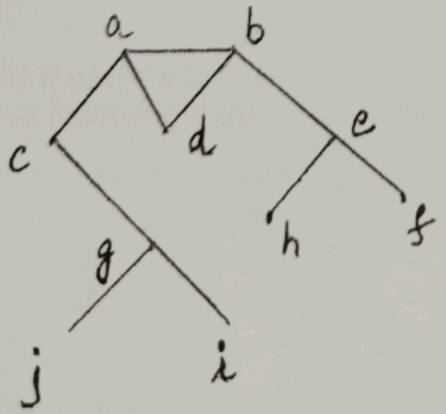
Q.NO	14	CO	5	CD	L3	PI	1.7.1	MARKS	5+5=10

a) Define prefix codes. Construct a binary tree for the given codes. A code for {a,e,n,r,t} is given by a:111, e=0, n= 1100, r:1101, t:10

Decode the sequence

11110111, 111110010, 1100010, 11010110010

b) Find the depth-first spanning tree for the graph shown in the figure if the vertex order is d,a,e,f,g,h,i,b,j,c



OR

Q.NO	15	CO	5	CD	L2	PI	1.7.1	MARKS	10

Define the following

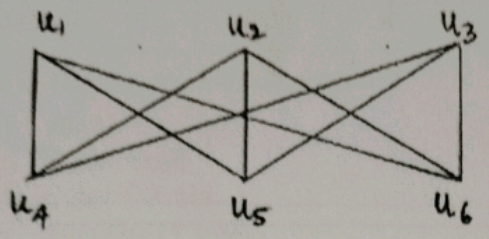
i) Spanning tree
ii) Binary tree
iii) Complete binary tree
iv) Weighted tree
v) Expression tree

Course Outcome (CO):	
CO–1	Use the knowledge needed to test the logic of the program.
CO–2	Demonstrate the concept of relations and be able to determine their properties.
CO–3	Explain and apply decode group codes based on the various methods
CO–4	Identify different types of graphs such as Isomorphism, Bipartite, Planar, Hamilton paths and cycles, Euler trails and circuits based on their properties.
CO–5	Explain different types of trees and apply tree traversal techniques like preorder and postorder.

PI	
1.2.1	Apply the knowledge of discrete structures, linear algebra, statistics and numerical techniques to solve problems.
1.7.1	Apply theory and principles of computer science and engineering to solve an engineering problem.
2.6.3	Identify existing solution/methods to solve the problem, including forming justified approximations and assumptions.

Cognitive Domain (CD):	
L1	Remember
L2	Understand
L3	Apply
L4	Analyse
L5	Evaluate
L6	Create

USN | | | | | | | | | | | | | Code:



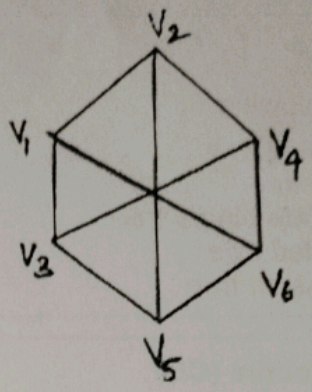


figure 1

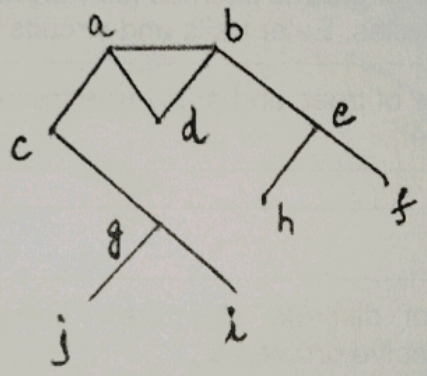


figure 2

-- End ---

USN	0	1	J	S	T	2	4	U	C	S	1	2	0	Course Code:	24CS340

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

III Semester B.E Degree Examination

Department of Computer Science and Engineering

COMPUTER ORGANIZATION

Duration: 3 Hours | Max. Marks: 100

Instructions:

- Answer all the questions.
- Part- B has internal choice.

PART – A

Q.NO	1	CO1	1	CD	L3	PI	2.2.3	MARKS	5+5

a. Explain the various registers available in the processor of a computer and describe how the processor is connected to memory.

b. What are condition code flags? Explain the four commonly used flags.

Q.NO	2	CO2	2	CD	L2	PI	1.4.1	MARKS	10

Describe in detail the interrupt mechanism? How can interrupts be enabled or disabled? Also, list the sequence of events involved in handling an interrupt request from a single device.

Q.NO	3	CO3	3	CD	L2	PI	1.4.1	MARKS	10

Discuss the implementation and working of static RAM and dynamic RAM cells, with suitable diagrams.

Q.NO	4	CO4	4	CD	L3	PI	2.2.3	MARKS	5+5

a. Explain the generate G_i and propagate P_i functions in a carry look-ahead adder for stage i, along with the bit-stage cell diagram.

b. Show the multiplication of 13 (multiplicand) and -11 (multiplier) using multiplier bit pair recoding technique.

Q.NO	5	CO5	5	CD	L2	PI	1.4.1	MARKS	5+5

a. List the actions needed to execute the instruction Load R5, X(R7).

b. Illustrate the conceptual hardware organization involving the register file and the ALU used for performing computations.

PART – B

Q.NO	6	CO1	1	CD	L3	PI	2.2.3	MARKS	5+5

a. Register R5 is used in a program to point to the top of a stack containing 32-bit numbers. Write a sequence of instructions using the Index, Autoincrement, and Autodecrement addressing modes to perform each of the following tasks:
i) Pop the top two items off the stack, add them, then push the result onto the stack.
ii) Copy the fifth item from the top into register R3.
iii) Remove the top ten items from the stack.
For each case, assume that the stack contains ten or more elements.

b. Explain the process of assembling and executing an assembly language program.

OR

Q.NO	7	CO1	1	CD	L3	PI	2.2.3	MARKS	5+5

a. Represent the decimal values 43, -10 and -27 in the following 7-bit binary formats.
i) sign-and-magnitude ii) 1's complement iii) 2's complement

b. What are stack frames? Discuss their role in the implementation of subroutines.

Q.NO	8	CO2	2	CD	L2	PI	1.4.1	MARKS	10

How does program-controlled I/O operate and why does it require continuous processor monitoring of I/O devices? Explain with a keyboard-to-display data transfer example.

OR

Q.NO	9	CO2	2	CD	L2	PI	1.4.1	MARKS	4+6

a. Describe the processor control registers that support interrupt processing.

b. Explain the timing of an input transfer on an asynchronous bus with a timing diagram.

Q.NO	10	CO3	3	CD	L2	PI	1.4.1	MARKS	10

Explain the organization of cache memory using fully associative mapping with a neat illustration. Also discuss how it improves the memory access time and how it is different from set associative mapping?

OR

Q.NO	*11*	*CO3*	3	CD	L3	PI	2.2.3	MARKS	5+5

a. *Consider a memory consisting of 64K words of 8 bits each. Discuss how this memory can be organized using 16K X 1 static memory chips.*

b. *With a block diagram, explain the virtual memory organization. Also list the benefits of a virtual memory system.*

Q.NO	12	CO4	4	CD	L3	PI	2.2.3	MARKS	5+5
a. Divide 1110 by 0011 using restoring division. b. Explain the IEEE standards for floating point number representation.									
OR									
Q.NO	13	CO4	4	CD	L2	PI	1.4.1	MARKS	10
With a neat diagram, explain the floating point addition/subtraction unit.									

Q.NO	14	CO5	5	CD	L2	PI	1.4.1	MARKS	10
Discuss the 5-stage processor datapath with the help of a neat diagram.									
OR									
Q.NO	15	CO5	5	CD	L2	PI	1.4.1	MARKS	10
With a neat diagram, explain the hardwired control unit.									

Course Outcome (CO):	
CO–1	*Apply the basic concepts of computer, instruction set architecture, function and characteristics of a computer system and design the assembly language for the sequence of machine instruction.*
CO–2	*Explain the different ways of communicating with the I/O devices.*
CO–3	*Design and analyse the performance of memory systems.*
CO–4	*Apply the Arithmetic and Logical operations with integer and floating–point operands.*
CO–5	*Illustrate the basic processing unit and organization of simple processor.*

PI	
1.4.1	*Apply theory and principles of computer science and engineering to solve an engineering problem.*
2.2.3	Identify existing solution/methods to solve the problem, including forming justified approximations and assumptions.

Cognitive Domain (CD):	
L1	Remember
L2	Understand
L3	Apply

--- End ---

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

VII Semester BE Degree Examination

Department of Humanities

UNIVERSAL HUMAN VALUES (UHV)-II

Duration: 2 Hours | Max. Marks: 60

Instructions:

- Answer all the questions in part-A and Part-B has internal choice.
- Indicate the question numbers correctly.

PART – A

Q.NO	1	CO	1	CD	L2	PI	8.1.1	MARKS	10
Define self exploration. What is the content of self-exploration?									
Q.NO	2	CO	2	CD	L2	PI	7.1.2	MARKS	10
What are the programs for ensuring the health of the body? Explain.									
Q.NO	3	CO	3	CD	L2	PI	7.1.2	MARKS	10
Explain the nine universal values (feelings) for understanding values in human-human relationship.									

PART – B

Q.NO	4	CO	4	CD	L2	PI	8.1.1	MARKS	10
Briefly explain the natural characteristics of the four orders in nature.									
OR									
Q.NO	5	CO	4	CD	L2	PI	8.1.1	MARKS	10
What is holistic technology? List all the criteria of holistic technology.									
Q.NO	6	CO	5	CD	L2	PI	7.1.2	MARKS	10
What do you mean by sustainable development? Discuss.									
OR									

USN												Code:	

Q.NO	7	CO	5	CD	L2	PI	7.1.2	MARKS	10
What do you mean by competence in professional ethics? Elaborate with examples.									

Q.NO	8	CO	1	CD	L2	PI	8.1.1	MARKS	10
Define value education and discuss the need for it in today's society.									
OR									
Q.NO	9	CO	1	CD	L2	PI	8.1.1	MARKS	10
Explain the difference between Human consciousness and animal consciousness with example.									

Course Outcome (CO): After completion of this course, student would be able to:	
CO–1	Understand self (I) clearly through self-exploration and demonstrate natural acceptance in situations by right understanding.
CO–2	Distinguish between self (I) body, and co-existence of both to be in harmony.
CO–3	Relate to family and society by understanding nine universal human values in human-human relationship for harmony and prosperity.
CO–4	Explain interconnectedness and mutual fulfilment at all four orders of nature for co-existence.
CO–5	Develop commitment for holistic understanding of harmony by natural acceptance of human values in his/her profession and society.

PI's	
7.1.2	Understand the relationship between the technical, socioeconomic and environmental dimensions of sustainability.
8.1.1	Identify situations of unethical professional conduct & propose ethical alternatives.

Cognitive Domain (CD):	
L1	Remember
L2	Understand
L3	Apply
L4	Analyse
L5	Evaluate
L6	Create

--- End ---